Lesson 2-8 · Session 1

Finding the Mean

Name _____ Date _____ Period _____

Maryland Standard 6.DS.4 · Learning target — I can find the mean of a data set by sharing the total equally among the values.
Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- The mean (average) equals the sum of all values divided by the number of values.
- The mean can be affected a lot by very high or very low values.
- Watch out — every value counts, including repeats and zeros.
- Check it — the mean must land between the least and the greatest value.

WORKED EXAMPLE

Finding the Mean of a Data Set Five afternoons of dog walking earned 14, 21, 17, 19, and 24 dollars. Find the mean.

Step 1 Add every value. The five amounts give a total of 95 dollars.

Step 2 Count the values. There are five of them.

Step 3 Share the total evenly: divide it by the count of values.

Step 4 Check: the mean should land between the least and the greatest value.

Answer The mean is 19 dollars — what each afternoon would earn if the money were shared evenly.

Practice

1. Find the mean of: 4, 8, 6, 10, 7

Sum _____ divided by _____ values Mean _____

Answer _____

SHOW YOUR THINKING

2. Five test scores are 82, 90, 76, 88, 94. Find the mean score.

Answer _____

SHOW YOUR THINKING

3. If one more score of 100 is added to problem 2's data (now 6 scores), will the mean go up or down? Explain without calculating.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

4. Find the mean of each set.

a. 3, 5, 7 _____
b. 10, 14, 12, 8 _____
c. 20, 25, 30 _____
d. 6, 6, 9, 11 _____
e. 40, 50, 45, 65 _____

Answer _____

SHOW YOUR THINKING

5. Ticket Money. Five families paid $18, $24, $15, $21, and $22 for tickets. Find the mean amount.

Answer _____

SHOW YOUR THINKING

6. Plant Watering. Six plants were given 12, 9, 15, 11, 14, 17 milliliters of water. Find the mean amount and explain what it means.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

7. One Big Value. Four students collected 8, 10, 12, 50 cans. Find the mean and explain why it sits above three of the four values.
Sentence Frame: "I know this because..."

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

8. Must the Mean Appear? Amari says the mean of a data set must always be one of the values in the set. Give a data set that shows he is wrong and explain.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

9. Find the mean of 11, 15, 19, and 23.

Answer _____ mean

SHOW YOUR THINKING

10. Find the mean of 7, 12, 9, 14, 18, and 12.

Answer _____ mean

SHOW YOUR THINKING

11. Test Average. A student's five test scores total 410 points. Find the mean score.

Answer _____ points

SHOW YOUR THINKING

12. Find the Mistake. Kai found the mean of 5, 9, and 13 by dividing 27 by 13. Explain the mistake and give the correct mean.

Answer _____ the mistake and the mean

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 2-7 Session 2. Find 9.9 ÷ 0.9.

Answer _____

SHOW YOUR THINKING

14. From Lesson 2-5 Session 1. Find the range of 14, 39, 22, 31.

Answer _____

SHOW YOUR THINKING

15. From Lesson 2-3 Session 2. Find the median of 10, 16, 22, 28.

Answer _____

SHOW YOUR THINKING

Test practice

16. Find the mean of 3, 7, 11.

□ A. 7	□ B. 6
□ C. 21	□ D. 11

17. Four values total 96. The mean is —

□ A. 26	□ B. 24
□ C. 384	□ D. 92

18. The mean is sometimes called the —

□ A. range	□ B. median
□ C. average	□ D. mode

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Include every data value, including zero, in the count. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Check one response against the data. State which measure or feature supports your answer.